# ALEXANDER ANDERSON

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Electrical engineer with experience building end-to-end medical-device data software, automated IC test systems, and robotics controls. B.S. Electrical Engineering with an Artificial Intelligence minor.

## TECHNICAL SKILLS

Software & Data: Python, SQL, Snowflake, dbt, Flask REST APIs, Streamlit, OpenCV, Docker, Git, Linux, automated testing

Electrical & Embedded: C/C++, C#, MATLAB, ROS2, Jetson, PXIe, NI LabWindows/CVI, PCB design, KiCad, Fusion 360, UART/I²C/CAN, oscilloscopes, DAQ

AI & Signal Processing: Snowflake Cortex, AI_EXTRACT, neural networks, reinforcement learning, A* search, fMRI/MRI processing

## PROFESSIONAL ENGINEERING EXPERIENCE

**Engineering Intern — Neuro Clinical Technology** | BIOTRONIK, Lake Oswego, OR | Jun–Sep 2026

- Designed and built BioSleep, an end-to-end sleep-algorithm characterization and testing platform: Snowflake/dbt database and pipelines (152 dbt tests passing), Flask REST API (~43 routes; 1,268 unit tests), and Streamlit-in-Snowflake review interface.
- Reverse-engineered device actigraphy encoding and developed decoding/ingestion workflows; joined implant, wrist-wearable, Apple Health, and handwritten diary data onto common nightly and epoch timelines with identity masking and reviewer corrections.
- Automated paper-diary extraction with Snowflake AI_EXTRACT and OpenCV mark classification, achieving 91% overall accuracy and 98% on clean marks across 70 hand-verified cells.
- Reduced degenerate-night flags from 70% to ~25% by resolving ingestion/non-wear issues; recovered ~82 device-days of activity history and diagnosed clock offsets up to ±7 hours (cross-source agreement 0.657→0.714 after correction).
- Replayed 59 debug nights and characterized 60 algorithm settings across 89 nights; created a citation-grounded Snowflake Cortex assistant, authored 10 runbooks, and presented results to the clinical technology team.

**Electrical Engineering Intern — Quality & Reliability** | Microsystems Engineering Inc., Portland, OR | May–Aug 2024

- Developed C/C# and NI LabWindows/CVI software for automated electrical testing of implantable medical-device ICs; integrated NI PXIe-4145/4163 source-measure units, PXIe-6570 digital I/O, PXIe-2567 relay matrix, and DAQ hardware.
- Built a Visual Studio instrument-control GUI and programmed automated continuity, isolation, current, and voltage test sequences with relay switching, safe power sequencing, measurement thresholds, and fault detection.
- Debugged PXIe timing, relay mapping, and hardware/software integration; documented test setup and troubleshooting procedures and supported IC packaging, wire-bond, and pad-classification analysis.

**Lead Controls Engineer — NASA Lunabotics** | University of Portland Robotics (UPR) | Sep 2025–May 2026

- Led controls work for lunar-regolith robot: manual joystick/keyboard teleoperation, gradual motor acceleration/deceleration, sensor integration, and ROS2-based control workflows.
- Integrated Jetson-to-microcontroller UART/CAN links, multi-camera telemetry/CSV logging, current sensing, and 24 V power distribution; designed dust-resistant electronics enclosure with gaskets, cable glands, and printed mounts.
- Contributed A* path planning and AprilTag localization; produced schematics, wiring diagrams, and CAD for NASA SRR/PDR/CDR technical reviews.

## BIOMEDICAL RESEARCH & ENGINEERING PROJECTS

**Undergraduate Student Researcher — LBNP Chamber** | Dr. Kimball Biomedical Lab, University of Portland | Jun–Sep 2023

- Designed and fabricated a full-scale lower-body negative-pressure (LBNP) research chamber: sliding acrylic half-cylinder, exam-table rail system, neoprene seal, and vacuum-pump interface.
- Designed a custom Fusion 360 PCB for pressure sensors, microcontroller/DAQ interfacing, and safety interlock; integrated pressure instrumentation and emergency cutoff relay.
- Prepared research proposal and safety protocols addressing orthostatic stress and physiological countermeasures.

**Student Engineer — Medical Equipment Repair** | Engineering World Health, Dominican Republic | Jul–Aug 2022

- Collaborated with hospital staff to diagnose, repair, and document infusion pumps, electrosurgical units, oxygen concentrators, and vital-sign monitors in resource-constrained clinical settings.
- Performed board-level troubleshooting with multimeters and oscilloscopes, component replacement, and equipment calibration; prepared bilingual repair logs and maintenance guidance.

## SELECTED ACADEMIC AND TECHNICAL PROJECTS

**MRI/fMRI and Biomedical Signal Processing** | University of Portland | Aug–Dec 2025

- Used SPM12, MATLAB, EEGLAB, and Python for motion correction, co-registration, normalization, smoothing, statistical contrasts, BOLD response analysis, and activation mapping.

**Artificial Intelligence & Robotics** | University of Portland | 2025

- Implemented A*, minimax, neural-network and reinforcement-learning agents; developed Python/OpenCV perception, localization, state-machine control, and robotic sensing workflows.

**Verilog Digital Systems Modeling** | University of Portland | Aug–Dec 2023

- Designed and simulated modular Verilog HDL circuits using behavioral, dataflow, and gate-level descriptions; developed testbenches and analyzed waveforms.

**Embedded Systems Design** | University of Portland | Aug–Dec 2023

- Built microcontroller-based embedded designs with timers, interrupts, ADC, UART/$I^2C$ interfaces, and hardware/software debugging.

## EDUCATION

**University of Portland — B.S. Electrical Engineering; Minor in Artificial Intelligence | May 2026**

Electrical Engineering GPA: 3.15 | AI Minor GPA: 3.5 | Merit scholarship recipient

## LEADERSHIP & ADDITIONAL EXPERIENCE

- IEEE Student Chapter, University of Portland — Treasurer/Secretary (2022–2024); member.
- Phi Theta Kappa Honor Society, Portland Community College — regional and chapter leadership; Dean's List.
- Additional hands-on experience in fabrication, construction, and electromechanical assembly.